Unit Plan: Chemistry Basics

FOR THE TEACHER

Summary

The AACT High School Classroom Resource library and multimedia collection has everything you need to put together a unit plan for your classroom: lessons, activities, labs, projects, videos, simulations, and animations. We searched through our resource library and constructed a unit plan for introducing the basic chemistry concepts to students: Laboratory Safety, Equipment, and Reports, Periodic Table Basics, Physical and Chemical Properties and Changes, Endothermic and Exothermic Changes, and Classification of Matter. These topics are very important for your students to master before they dig into other chemistry concepts. This unit is designed to be used at beginning of the school year.

Grade Level

High School

NGSS Alignment

The teaching resources used in this unit plan will help prepare your students to meet the performance expectations in the following standards:

- **HS-PS1-5:** Apply scientific principles and evidence to provide an explanation about the effects of changing the temperature or concentration of the reacting particles on the rate at which a reaction occurs.
- **HS-PS2-6:** Communicate scientific and technical information about why the molecular-level structure is important in the functioning of designed materials.
- **HS-PS3-2:** Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles (objects) and energy associated with the relative positions of particles (objects).
- **HS-ETS1-3:** Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics as well as possible social, cultural, and environmental impacts.
- **Scientific and Engineering Practices**:
    - Analyzing and Interpreting Data
    - Asking Questions and Defining Problems
    - Planning and Carrying Out Investigations
    - Engaging in Argument from Evidence
    - Obtaining, Evaluating, and Communicating Information
    - Developing and Using Models
    - Constructing Explanations and Designing Solutions

Objectives

By the end of this unit, students should be able to

- Distinguish between safe and unsafe behavior in the chemistry laboratory.
- Understand the importance of following safety rules in a chemistry laboratory.
- Responsibly follow safety guidelines presented in a chemistry laboratory.
- Correctly identify and name common pieces of laboratory equipment.
- Associate a hazard symbol with its meaning.
- Understand the importance of hazard symbols.
- Identify relevant, independent, dependent, and control variables for a research question.
- Plan and organize the steps necessary to conduct a simple investigation.
- Write a formal lab write up.
- Identify important features and characteristics of good graphs.
- Identify positive and negative correlations from a graph.
- Interpolate and extrapolate based on the patterns presented in a graph.

- Generate a well-formatted graph to present a set of data.
- Determine a method to measure the mass and volume of an irregular object.
- Accurately use laboratory equipment to gather data.
- Calculate the density of an irregular object using their data.
- Create a graph of mass vs. volume using class data and use the slope of the line to calculate the average density of the objects.
- Create a lab report using tables and graphs, following a provided template.
- Explain how the accuracy of a measurement will change depending upon the measuring tool used to measure.
- Determine the correct measurement based on the markings on the device used.
- Identify the uncertainty value for a measurement based on the markings on a measurement device used.
- Distinguish between the states of matter at the particle level.
- Explain, using examples how matter is different in one state versus another.
- Identify examples of different states of matter.
- Classify the three states of matter found in the laboratory by molecular level particle representations.
- Identify differences in the particle representations to classify them as pure substances, both elements and compounds, as well as mixtures.
- Verbally explain the classification system their group developed.
- Understand vocabulary related to chemistry.
- Identify whether a physical or chemical change has occurred.
- Provide evidence supporting which change has occurred.
- Identify physical properties substances.
- Identify appropriate methods for separating mixtures.
- Better understand how the periodic table is organized.
- Classify elements by family name, group number and period number.
- Identify reactions as either endothermic or exothermic.

Chemistry Topics

This unit supports students' understanding of

- Laboratory Safety
- Laboratory Equipment
- Data collection
- Graphing and analyzing data
- Quantitative Chemistry
- Measurement
- SI Units
- Significant Figures
- Uncertainty
- States of Matter
- Molecular Motion
- Physical Properties
- States of matter
- Pure substances
- Mixtures
- Separating mixtures
- Elements
- Periodic Table
- History

- Evidence of a chemical reaction
- Physical changes and properties
- Chemical changes and properties
- Endothermic/exothermic reactions

Time

Teacher Preparation: See individual resources.
Lesson: 10 - 15 class periods, depending on class level.

Materials

- Refer to the materials list given with each individual activity.

Safety

- Refer to the safety instructions given with each individual activity.

Teacher Notes

- This unit plan begins with an introduction to the laboratory that includes safety guidelines, equipment, and lab reports. It then moves through activities that introduce other chemistry basics topics. You can use this unit plan as written or change the order to meet the needs of your students.
- The teacher notes, student handouts, and additional materials can be accessed on the page for each individual activity.
- Please note that most of these resources are AACT member benefits.
- Laboratory safety is an important topic for teachers of chemistry, many of whom may not have had access to enough training. Before planning your lab activities and demos, *read* *Keeping the Wow Factor, and Controlling the Risks*, an article from the September 2016 issue of *Chemistry Solutions*. This article reviews the safety problems inherent in the "traditional" rainbow experiment and similar demonstrations, includes the responses from various organizations, and also provides safer alternatives.
- Another article that you might find helpful is *There's More to the New Safety Data Sheets than a Missing "M"* from the May 2017 issue. This article aims to increase your comfort level with the new SDSs by describing the timeline, some changes over the past 5 years, the pros and cons of the newer format, related hazard communication issues, and information on other available resources.
- Finally, if you're unsure of what to do with laboratory waste, read *Managing Chemical Wastes in the High School Lab* from the May 2016 issue. This article provides a solid starting point to determine proper disposal methods for high school lab waste and includes a *Quick Disposal Reference Guide* that you can download, print, and hang in your chemical prep room.
- Teachers should refer to the ACS published document, *Guidelines for Chemical Laboratory Safety on Secondary Schools* as a great resource for teaching lab safety in the chemistry classroom. Additional specific information that could be used when teaching hazard symbols can be found starting on page 21.

Classroom Resources:

Laboratory Safety, Equipment, and Reports

- Use one or more of the videos from the ACS Chemical Safety video series to introduce important safety concepts to your students. This series includes five student videos and one video for teacher use. All of the videos are unlocked so that your students can access them. The RAMP lesson plan includes several activities to help students become more knowledgeable and better able to assess risks and hazards in the lab. Student videos include:
    - Safety Mindset
    - Safety Data Sheets (SDS)

    - How to Dress for the Lab and Personal Protective Equipment
    - Preparing for Emergencies
    - RAMP for Students

- The final video, RAMP for Teachers, outlines steps you can take to make sure your students are as safe as possible while exploring and experimenting in the lab.
- Use the activity, What Not to do in the Chemistry Lab to introduce laboratory safety and best practices and discuss these important topics with your students. During the activity students examine a cartoon of a chaotic chemistry lab and note the specific behaviors that are dangerous and unsafe in a chemistry laboratory setting.
- The Hazard Symbols activity from the May 2018 issue of *Chemistry Solutions* is a great way to familiarize your students with common hazard symbols and their meaning. As optional extensions to this activity, use chemical containers that contain these hazard symbols as examples for your students. You could also share an example of a MSDS with students as part of a discussion about these hazard symbols.
- Then use the activity, Analyzing & Creating Safety Labels to further help your students understand the color and symbols on the Safety Diamond and apply their knowledge to interpreting a chemical label.
    - Another option is the Lab Safety and Safety Data Sheets (SDS) activity that helps students learn how to identify safe lab practices with a focus on labeling and background safety information for reagents used in the lab.
- Experimental design can help students prepare for labs where they may be writing their own procedures or even need to decide on a question to test a phenomenon. The activity, Procedure Pictures introduces the idea of generating questions and testing those questions.
- Many chemistry students come to the class with a limited understanding of how to write high quality lab reports. Before your students start working in the lab, read Tools and Strategies for Teaching Lab Report Writing from the September 2016 issue of *Chemistry Solutions* for some ideas that will help your students with their reports. This article includes an accompanying lab, Investigating the Density of an Irregular Solid Object. You may want to use the Density Animation to introduce this concept to your students.
- Graphing is an essential skill for creating a detailed lab report. The Graphing Simulation includes an accompanying activity to teach students graphing technique and data analysis.
- The article, *Cleaning Up the Lab* from the March 2022 issue of *Chemistry Solutions* has insights on creating a procedure to teach students to share the responsibility of making sure the lab stays clean. This can alleviate your mental load and make lab days run more smoothly.
- Use the lesson plan, How To Write a Formal Lab to teach students how to put the parts of a formal lab report together. Having students familiarize themselves with this format will expedite teacher grading.
- Introduce your students to basic lab equipment with the activity, The Essentials for Survival. In addition to learning about equipment, students will model appropriate group work, class discussions and practice wiring efficient Claim-Evidence-Reasoning (CER) reports.
    - The activity, Race to Safe, is a great follow-up here. Students will be presented with a variety of lab safety situations and asked how to handle them in the lab. The lab involves them showing the response to each situation, incorporating fun, and kinesthetic activity to improve memory.
- The Laboratory Equipment Memory Game published in the March 2018 issue of *Chemistry Solutions* is a fun and effective activity to assess your students' understanding of common lab equipment. This resource includes a card template that you can use to print the game cards on stock paper. You can reuse the cards from year to year by laminating them and then storing them in zip-lock bags.
    - Then use the Laboratory Equipment Bingo activity to see how well your students know common equipment that can be found in a chemistry lab. This activity includes 32

unique bingo cards and a PowerPoint presentation with pictures of 49 pieces of lab equipment.

Introductory Topics

- You may want to start off the year showing one of the videos from our multimedia library.
    - The Ancient Chemistry Video traces the history of chemistry from the discovery of fire, through the various metal ages, and finally to the great philosophers. This resource includes an activity sheet for students to use while viewing the video.
    - You might prefer to pique your students' interest in chemistry with the Arsenic Video and hear Sam Kean, author of *The Disappearing Spoon,* tell stories about arsenic, a deadly element that was once referred to as the "Inheritance Powder". Additional videos about the elements can be found in our multimedia library.
    - Another option is the Frontiers of Chemistry video that explores new scientific developments made possible by the application of fundamental chemistry concepts. This video also includes a student activity sheet.
- Students explore the properties of the states of matter with the Categorizing States of Matter activity, which has them analyze both written statements and images that describe the properties of a solid, liquid or gas. Students then determine which state of matter the description best describes and categorize it accordingly.
- The Visualizing States of Matter activity has students view, sort and classify pure substances and mixtures into the 3 common states of matter found in the laboratory. They then discuss their classification system with their teacher and peers.
- Assess your students' understanding of the concept with a 10 question quiz in the States of Matter and Phase Changes simulation. This activity challenges students to identify the correct state of matter and connect it with an animated particle diagram.
- In the Chemical and Physical Changes Lab students observe and analyze a number of interactions and determine if a chemical or physical change occurred. This activity will help students understand vocabulary related to chemistry, identify whether a physical or chemical change has occurred, and provide evidence supporting which change has occurred.
- The Classifying Matter Animation will help students become familiar with definitions and examples of several broad classifications of matter, including pure substances (elements and compounds) and mixtures (homogeneous and heterogeneous). The animation includes real-life examples as well as particles for each type of matter.
    - Follow up with the activity, Elements, Compounds, & Mixtures – Oh My! to provide students with more practice with classification of matter and particle diagrams.
    - Then use the Separation of a Mixture Lab to allow students to devise their own method to separate a mixture of sand, salt, poppy seeds, and iron filings after they identify the physical properties of each and identify appropriate methods for separating them.
- Introduce the Periodic Table to your students with The Scavenger Hunt activity from the PTable.com Investigations Activity. This lesson walks students through an investigation of a large number of topics, from physical properties to history of the elements.
    - A second option is to use the video, How the Periodic Table Organizes the Elements from the American Chemical Society to introduce the basics to your students. The resource includes a student activity sheet for students to answer questions during the video and an answer key.
- Students can investigate physical and chemical changes using the lab, Determining a Physical or Chemical Change. They will appreciate the color changes and apply their knowledge.
- The lab, Chemistry in a Bag will help students distinguish between chemical and physical changes, identify evidence of a chemical change, and understand the difference between exothermic and endothermic reactions. Students also use this lab to understand the Law of Conservation of Mass.

- Finish up the unit with the Chemistry Basics Advanced Crossword Puzzle to assess your students' understanding of fundamental chemistry topics. There is also a version of the puzzle written for middle school students.

Culminating Activities

- If time permits, you may want to use one of the following resources as a culminating event for this introductory unit.
    - In the activity, Lab Safety, You're Fired!, students read an account of a laboratory tour that details numerous safety infractions. They are then charged with identifying the safety violations and determining which scientist working in the lab should be fired.
    - Students research an actual industrial chemical accident in the project, Chemical Disasters: Good Chemicals Gone Bad! by examining the chemicals involved including uses, hazards, chemical and physical properties.